

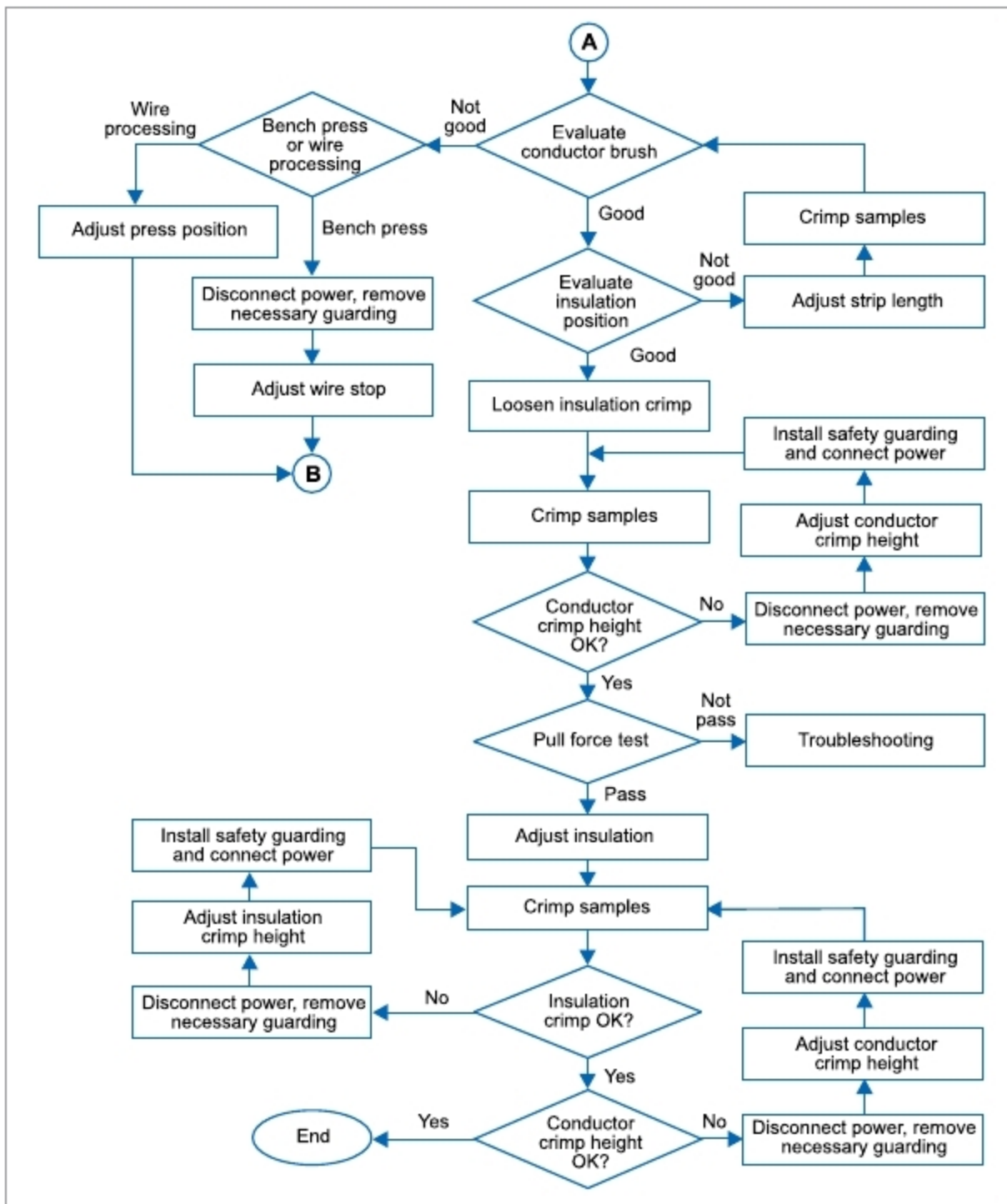


Fig. 1: Crimping process at a glance

which is made using hand tools or semi-automatic crimping machines.

Crimped terminals are inserted into moulded housing cavities with the help of contact insertions and removal tools. Then, complete crimped connectors are checked as per predefined standards and specifications.

The entire crimping process can be described step by step as follows (Fig. 1):

1. Check if tooling is clean and not worn out. If necessary, clean and replace damaged/worn out tooling.
2. Disconnect power to the press and remove guarding devices.
3. Install appropriate tooling into the press.
4. Load terminals into the tooling, so that the first terminal is located over the anvil.

5. Manually cycle the press to ensure a complete cycle without interference. If it cannot, remove tooling and check press shut height. Go to step 3.

6. Check if the tooling is aligned. Check the impression on the bottom of the crimp made by anvil tooling. Make sure extrusions and crimp forms are centred. If not, align tooling and go to step 5.

7. Check whether the terminal feed locates the next terminal over the centre of anvil tooling. If not, adjust terminal feed and feed finger, and go to step 5.

8. Re-install all safety devices that were removed during setup. (Follow all safety requirements listed in individual press and/or tooling manuals.)

9. Crimp sample terminals under power.

10. Evaluate cut-off tab length and conductor bell mouth. If adjustment is necessary, disconnect power to the press and remove guarding. Adjust track position. Manually cycle the press and check the feed finger for feed location. Go to step 7.

11. Evaluate conductor brush. If adjustment is necessary, disconnect power to the press and remove guarding. Adjust wire stop for bench applications, or press position on automatic wire-processing equipment. Go to step 8.

12. Evaluate insulation position. If necessary, adjust strip length, crimp new samples and go to step 11.

13. Loosen insulation crimp height.

14. Crimp sample terminals.

15. Measure conductor crimp height and compare with specifications. If necessary, disconnect power and remove guarding. Adjust conductor crimp height, install guards, connect power and go to step 14.

16. Perform pull force test. If it fails, refer to troubleshooting.

17. Adjust insulation crimp.

18. Crimp sample terminals.

19. Evaluate insulation crimp. If necessary, disconnect power and remove guarding. Adjust insulation crimp height, install guards, connect power and go to step 18.

20. Measure crimp height and compare with specifications. If necessary, disconnect power and remove guarding. Adjust conductor crimp height, install guards, connect power and go to step 18.

Good manufacturing practices

Pollution control. The government places utmost importance on controlling environmental pollution. Therefore small-scale entrepreneurs need to follow environmental-friendly practices and adopt pollution-control measures by process modification and technology substitution.

India acceded to Montreal Protocol in September 1992. Hence, production and use of ozone-depleting substances like chlorofluoro carbon (CFC), carbon tetrachloride, halons and methyl chloroform need to be phased out immediately with alternative chemicals/solvents. A notification